

SINONG (SIMON) ZHAN

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EDUCATION

Northwestern University

Sept 2023 - Present

PhD in Electrical and Computer Engineering, advised by Prof. Qi Zhu

Northwestern University

Sept 2023 - Jun 2026

Master in Computer Engineering, advised by Prof. Qi Zhu

University of California, Berkeley

August 2018 - December 2022

Bachelor of Arts in Computer Science and Applied Mathematics, advised by Prof. Sanjit Seshia

RESEARCH INTERESTS

My research lies at the intersection of **reinforcement learning**, **formal methods**, and **AI safety**. I design safe and delay-robust RL algorithms integrating formal verification, barrier certificates, constraint satisfaction, with learning-based control to provide provable guarantees for autonomous systems. More recently, I develop **specification-guided frameworks** that leverage formal specifications (temporal logics, program contracts) to systematically **evaluate**, **steer**, and **explain** foundation model-based agents, addressing the challenge of **bridging informal human safety intent and provably correct agent behavior**.

RESEARCH EXPERIENCE

IDEAS Lab, Northwestern University

Sep 2023 - Present

Graduate Research Assistant

Advised by Prof. Qi Zhu

- Developed safe and delay-robust reinforcement learning algorithms integrating formal verification (barrier certificates, constraint satisfaction) for various safety-critical robotic applications, published at ICML ×2, NeurIPS Spotlight, ICLR ×2, and L4DC ×2.
- Designed imitation learning frameworks using generative modeling for stable training process in stochastic environments, with applications to behavior modeling and reward shaping.
- Explored neuro-symbolic approaches combining large language models with formal reasoning to enhance interpretability and verifiability of autonomous systems.
- Built **SENTINEL**, a multi-level formal framework that evaluates Foundation Model based embodied agents against temporal logic specifications in realistic simulation environments.

University of California, Berkeley

Feb 2022 - April 2023

Undergraduate Researcher & Research Assistant

Advised by Prof. Sanjit Seshia

- Developed automated verification and control synthesis methods for complex robotics systems, specializing in path planning and trajectory optimization for underwater robots.
- Created SMT-based 3D bin-packing solver for design model checking, contributing to structured design automation pipelines and formal verification frameworks for NSF UUV Challenge.

Human Computing Lab, ISCAS & XDiscovery Lab, Dartmouth

May 2019 - Sep 2021

Research Assistant

Advised by Prof. Feng Tian, Prof. Teng Han, & Prof. Xing-Dong Yang

- Designed VR-based interaction models with real-time feedback systems and developed novel microfluidics-based sensing technologies, resulting in CHI and UbiComp publications.
- Implemented self-supervised learning techniques for robust multi-modal sensor fusion in haptic interfaces, exploring fast-prototyping methods for human-computer interaction applications.

INDUSTRY EXPERIENCE

Google

Research Intern

Start June 2026

Google Research & P+D Team

- Built end-to-end RL training infrastructure for long-horizon coding agents Google Xmanager Infra, scaling models post-training from 0.5B to 30B with tensor/sequence parallelism, ZeRO-1 distributed optimizer, and colocated rollout training memory management.
- Designed a summarization-based RL method for long-horizon coding agents in which the policy learns to select informative context from its own trajectory history while interleaving reasoning and tool calls, with a rollout and credit-assignment scheme that differentiates summarization, reasoning, and tool-call tokens rather than broadcasting a single trajectory-level advantage uniformly.

Amazon

Applied Scientist Intern

June 2025 - Sep 2025

Store Foundation AI Team

- Developed multi-turn conversational agents for Rufus Shopping Assistant using state-of-the-art reinforcement learning algorithms and designed logic-based reward machines for human behavior simulation.
- Contributed to large-scale production systems serving millions of Amazon customers, optimizing LLM agent training pipelines for e-commerce applications and customer interaction workflows.

ACADEMIC SERVICE

Conference Reviewer: NeurIPS, ICML, ICLR, AAAI, L4DC, ASP-DAC, CVPR, ECC, ECCV

Journal Reviewer: Machine Learning Journal (Springer), IEEE Internet of Things Journal

Program Committee: ICCPS Artifact Evaluation Committee

PUBLICATIONS

*Full list with PDFs at simon-zhan.com and Google Scholar. * = equal contribution.*

Selected Papers (Safety, RL, and Formal Methods)

S. S. Zhan, Q. Wu, F. Yang, X. Shi, C. Huang, Q. Zhu. “Belief-Based Offline RL for Delay-Robust Policy Optimization.” *ICLR*, 2026.

S. S. Zhan, P. Wang, Q. Wu, Y. Wang, R. Jiao, C. Huang, Q. Zhu. “Enhancing Inverse RL through Encoding Dynamic Information in Reward Shaping.” *L4DC*, 2026.

Q. Wu*, **S. S. Zhan***, Y. Wang*, Y. Wang, C.-W. Lin, C. Lv, Q. Zhu, J. Schmidhuber, C. Huang. “Directly Forecasting Belief for RL with Delays.” *ICML*, 2025.

Q. Wu*, **S. S. Zhan***, Y. Wang, Y. Wang, C.-W. Lin, C. Lv, Q. Zhu, C. Huang. “Variational Delayed Policy Optimization.” *NeurIPS*, 2024. **(Spotlight)**

S. S. Zhan, Y. Wang, Q. Wu, R. Jiao, C. Huang, Q. Zhu. “State-wise Safe RL with Pixel Observations.” *L4DC*, 2024.

Q. Wu, **S. S. Zhan**, Y. Wang, Y. Wang, C.-W. Lin, C. Lv, Q. Zhu, J. Schmidhuber, C. Huang. “Boosting RL with Strongly Delayed Feedback Through Auxiliary Short Delays.” *ICML*, 2024.

Y. Wang, **S. S. Zhan**, R. Jiao, Z. Wang, W. Jin, Z. Yang, Z. Wang, C. Huang, Q. Zhu. “Enforcing Hard Constraints with Soft Barriers: Safe RL in Unknown Stochastic Environments.” *ICML*, 2023.

Y. Wang*, **S. S. Zhan***, Z. Wang, C. Huang, Z. Wang, Z. Yang, Q. Zhu. “Joint Differentiable Optimization and Verification for Certified RL.” *ICCPS*, 2023.

F. Yang, **S. S. Zhan**, Y. Wang, C. Huang, Q. Zhu. “Runtime Safety Verification of Neural Network Controlled System.” *RV*, 2024.

H. Su, S. Feng, **S. S. Zhan**, N. Zhan. “Switching Controller Synthesis for Hybrid Systems Against STL Formulas.” *FM*, 2024.

Additional Papers

W. Feng, L. Wang, T. Wei, J. Zhang, C. Gao, **S. S. Zhan**, P. Lv, W. Dong. “Token Buncher: Shielding LLMs from Harmful Reinforcement Learning Fine-Tuning.” *CCS*, 2026.

X. Shi, J. Ding, X. Zhao, **S. Zhan**, et al. “STEP-LLM: Generating CAD STEP Models from Natural Language with LLMs.” *DATE*, 2026.

C. Li, W. Tang, Y. Huang, **S. S. Zhan**, M. Hu, X. Jia, Y. Liu. “Shedding Light on VLN Robustness: Indoor Lighting-based Adversarial Attack.” *CVPR*, 2026.

Y. Zhang, T. Wang, J. Gesi, Z. Wang, Y. Lu, J. Lin, **S. Zhan**, et al. “Shop-R1: Rewarding LLMs to Simulate Human Behavior in Online Shopping via RL.” *ICLR*, 2026.

Y. Zhang, Z. Wang, Y. Lu, **S. Zhan**, J. Huang, D. Wang. “See, Think, Act: Online Shopper Behavior Simulation with VLM Agents.” *SEA @ NeurIPS*, 2025.

Y. Wang, R. Jiao, C. Lang, **S. S. Zhan**, et al. “Empowering Autonomous Driving with LLMs: A Safety Perspective.” *LLMagent @ ICLR*, 2024.

R. Jiao*, Y. Wang*, X. Liu, **S. S. Zhan**, C. Huang, Q. Zhu. “Kinematics-aware Trajectory Generation and Prediction with Latent Stochastic Differential Modeling.” *IROS*, 2024.

W. Sun, Y. Chen, Y. Chen, **S. S. Zhan**, et al. “MicroFluID—Reconfigurable RFID Platform for Interaction Sensing via Microfluidics.” *UbiComp*, 2022.

W. Sun, Y. Chen, **S. S. Zhan**, T. Han, F. Tian, H. Wang, X.-D. Yang. “RElectrode: Reconfigurable Electrode for Multi-Purpose Sensing via Microfluidics.” *CHI*, 2021.

Under Review

S. S. Zhan, Y. Liu, P. Wang, Z. Wang, Q. Wang, Z. Ruan, X. Shi, X. Cao, F. Yang, K. Wang, H. Shao, M. Li, Q. Zhu. “**SENTINEL**: A Multi-Level Formal Framework for Safety Evaluation of Foundation Model-based Embodied Agents.”

S. S. Zhan*, Q. Wu*, A. Ruan, F. Yang, P. Wang, Y. Wang, R. Jiao, C. Huang, Q. Zhu. “Inverse Delayed Reinforcement Learning.”

Q. Wu, Y. Wang, **S. S. Zhan**, Q. Zhu, J. Schmidhuber, C. Huang. “Efficient Multi-Step RL with Expectation-Maximization Bootstrapping.”

Q. Wu, Y. Wang, **S. S. Zhan**, Y. Dai, S. Deng, S. Habchi, Q. Zhu, M. Gallé, C. Huang. “A Unified Framework for Rethinking Policy Divergence Measures in GRPO.”

Y. Peng*, P. Wang*, **S. S. Zhan***, Y. Lyu, Z. Ni, J. Yan, F. Wong, R. Jiao, H. Yin, X. Cao, H. Shao, M. Li, R. Zhang, Q. Zhu. “**ManiGuard**: A Benchmark and Data Suite for Specification-Grounded Safety Evaluation and Improvement of Robotic Manipulation.”

TEACHING EXPERIENCE

University of California, Berkeley
Teaching Assistant

Fall 2022

Math 128A: Numerical Analysis

- Conducted weekly discussion sections and office hours for 40+ undergraduate students
- Graded assignments and provided feedback on numerical methods and computational mathematics

PATENTS

EMG-based Multi-target Control System. Simon Zhan, Junjun Fan, Feng Tian, Wei Sun. Chinese Patent No. CN201710953534.X.

Microfluidic Pipeline Composite Structure and Pattern Deformation System. Wei Sun, Yanjun Chen, Simon Zhan, Teng Han, Feng Tian, Hongan Wang, Xing-Dong Yang. Chinese Patent No. CN202110377915.4.

Fluid Pattern Reconfigurable System Based on Microfluidic Technology. Wei Sun, Yanjun Chen, Simon Zhan, Teng Han, Feng Tian, Hongan Wang, Xing-Dong Yang. Chinese Patent No. CN202110378536.7.

TECHNICAL SKILLS	
Programming	Python, C/C++, Lean FRO, CUDA, Java, R, Julia
ML/AI Frameworks	PyTorch, JAX, Hugging Face, Verl
Optimization	Gurobi, CVX, CasADi
Tools	Git, AWS, Docker, L ^A T _E X, Linux, Unity3D, MATLAB, Simulink
Languages	English (Native), Chinese (Native), German (Limited-proficiency)